

HOW BAD COULD IT BE? ACCIDENT SEVERITY PREDICTION

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ABSTRACT. We analyze the relations between traffic infrastructure, light and weather conditions, and accident severity. In our research, we find that among light and weather conditions, air pressure and humidity play a major role in accident severity while light conditions play little role in this regard. Among infrastructure, we find that the presence of major stations and road crossings are key indicators for accident severity. A discussion of ethical considerations of the data and our conclusions is included.

1. RESEARCH QUESTION AND OVERVIEW OF THE DATA

Since the invention and popularization of the automobile, the traffic accident has found a place among the leading causes of death in the US—an accomplishment for which car manufacturers are to be congratulated. A recent report from the CDC found that, of 29 high income nations examined, the US suffers the most traffic deaths per capita, with a fatality rate about 2.3 times the average of the other 28 [3]. It also found that between 2015 and 2019, the fatality rate in the US increased by .1%, while it decreased an average of 10.4% in the other 28 nations. While it is discouraging that we have seen no improvement, we may look to the successes of other nations and work to emulate them. The CDC report found that an average of 36,791 people died in crashes each year in the US—101 each day—between 2015 and 2019. If we diminished traffic fatalities as other nations have, we could save over 3,000 lives each year—about ten lives every day. We have a moral imperative to change.

Alas, since there is little sign of the car going anywhere anytime soon, the most that we can hope to do is to account for frequent contributing factors to crashes. In this paper, we evaluate the impact of certain light and weather conditions and certain infrastructural features on the severity of accidents, as gauged by the length of road impacted by the accident and by a simple ordinal measure of severity provided in our dataset, which we drew from Moosavi et al. [1, 2]. Our dataset contains data of about 7 million car accidents in the United States. Relevantly, it captures several measures of the weather conditions at the time and place of each crash, whether each crash was near one or more of a certain infrastructure features, as well as

the length of road affected by the crash and a measure of “severity”—which seems to relate to the delay caused by the crash, but is not exclusively described by it. As discussed later, pedestrian crossings relate highly to the severity of an accident, which suggests it is also related to severity of injury. It may also be asked what the compilers of this dataset meant by “near.” Whether “near” means 5 feet, 50 feet, or 500 feet is not clear from the documentation of the dataset.

In our analysis provided here, we seek answers to questions such as, (1) Which factors have the greatest impact on accident severity? (2) Which weather conditions correlate with more severe accidents? (3) How does traffic infrastructure impact accident severity? (4) Which factors, in combination, compound accident severity?

2. DATA CLEANING / FEATURE ENGINEERING

Some of the features in the data presented challenges for interpretation, including city, state, and county names as well as wind direction and airport codes. These, and other similar features, were removed to make analysis easier. Furthermore, some of the data entries were missing some feature data. This was handled using two different methods: removal of data and averaging of data. In one group we removed any entries with missing data, which resulted in reducing our 7-million-entry dataset to around 5 million entries. This preserved data purity, but lost around 30% of the data. In the second group we replaced missing data with the averages of the features to preserve as much data as possible, but it is unclear whether replacing it with averages provides a good method to produce accurate results. In either case, we believe that our two methods of data cleaning produces a good representative of the dataset as a whole, preserving both aspects of purity and accuracy as well as can be done.

3. DATA VISUALIZATION AND BASIC ANALYSIS

An explanation of the features:

There are many features included in this dataset. Figure 1 outlines a description of some of the road infrastructure features that are referenced in the dataset.

Traffic Feature Frequency:

To begin our analysis, we consider the relative frequencies of each traffic feature in our data set in hopes of determining features that largely correlate with accident occurrence (See Figure 1). What this graphic suggests is that accidents occur most frequently near traffic signals, crossings, and junctions, suggesting that these infrastructure features largely contribute to accident occurrences.

Traffic Feature Co-Occurrence

To better understand the relation between each feature, we created a co-occurrence matrix to examine the proportion of accidents near each feature

Table 2: Definition of Point-Of-Interest (POI) annotation tags based on Open Street Map (OSM).

Type	Description
Amenity	Refers to particular places such as restaurant, library, college, bar, etc.
Bump	Refers to speed bump or hump to reduce the speed.
Crossing	Refers to any crossing across roads for pedestrians, cyclists, etc.
Give-way	A sign on road which shows priority of passing.
Junction	Refers to any highway ramp, exit, or entrance.
No-exit	Indicates there is no possibility to travel further by any transport mode along a formal path or route.
Railway	Indicates the presence of railways.
Roundabout	Refers to a circular road junction.
Station	Refers to public transportation station (bus, metro, etc.).
Stop	Refers to stop sign.
Traffic Calming	Refers to any means for slowing down traffic speed.
Traffic Signal	Refers to traffic signal on intersections.
Turning Loop	Indicates a widened area of a highway with a non-traversable island for turning around.

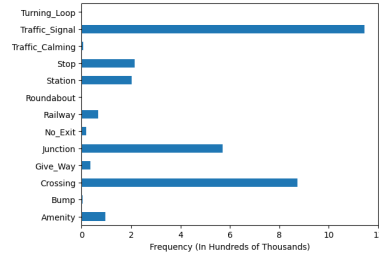


FIGURE 1. A brief description of infrastructure features and relative frequencies. Source: Moosavi et al, Accident Risk Prediction based on Heterogeneous Sparse Data: New Dataset and Insights

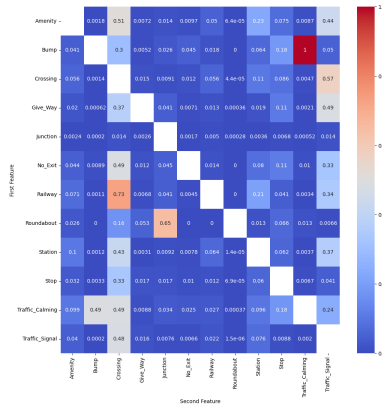


FIGURE 2. Co-Correlation Matrix of Traffic Features

that are also near each of the other features (See Figure 2). This visual reveals that crossings contribute a great deal to accident occurrences and their severity, especially in combination with other features. This indicates crossings as a highly-dangerous feature to drivers and pedestrians alike.

4. LEARNING ALGORITHMS AND IN-DEPTH ANALYSIS

In seeking to understand the impact of the various features on the severity of accidents and the distance of the road affected by accidents, we applied several machine learning models, including decision trees and random forests. We applied classifier machine learning methods with severity as the target data using classification accuracy as our metric, and we used regression methods with distance affected as the target data using mean-squared-error as our metric.

We first considered the impact of weather and daytime conditions on both the severity of and distance of traffic affected by accidents, including features of weather indicators as well as whether or not it was daytime during the

accident. In performing this analysis, we hoped to answer the question of which weather conditions contribute the most to the impact of accidents.

We applied both a decision tree classifier and XGBoost classifier to this weather-filtered data on accident severity. An unoptimized decision tree produced results with 81.13% accuracy when tested on test data, and an unoptimized XGBoost classifier produced results with 85.31% accuracy. XGBoost gave wind chill as the most important feature, followed by temperature and air pressure, while the decision tree classifier suggested air pressure as most important, followed by humidity and visibility. Surprisingly, the features indicating whether the accident occurred at night or day had little impact on predicting accident severity. This shows that the greatest indicator of traffic severity relies on the atmospheric conditions of the area, including air pressure, temperature, and wind. This result makes sense physically, as changes in the weather and increased wind speeds can cause unsafe driving conditions. While these weather conditions can compound to increase the severity of the accident, they can also prevent emergency personnel from quickly localizing to the scene of the accident. We further applied a decision tree regressor and XGBoost regressor to this filtered data. The decision tree produced predictions with a mean-squared-error of approximately 4.47, and the XGBoost produced predictions with a mean-squared error of approximately 3.24. While these may seem like large errors, it is relatively small in a dataset of about 5 million entries and a target range of 442 miles.

However, these MSEs and classification rates do not tell the full story. Across the 5 million plus data points, approximately 85% of them are given a severity rating of 2. This value is alarmingly close to the prediction accuracies of both classification models. Similarly, 75% of the distances recorded are less than 1.2 miles. Any model could experience reasonable success simply by guessing that the distance that traffic was impacted was 1 mile.

Fitting an unoptimized random forest classifier to a training subset of our infrastructure-filtered data, with temporal severity as our target, predicted the severity of the test subset with a success rate of 85.25% and identified crossings as the most important feature, followed by traffic signals, stops, and stations. Applying a Random Forest Regression to the same data, with the distance impacted as the target, yielded a mean squared error of 3.34 and identified traffic signals as most important, followed by crossings, stations, and stops. In our data, about 12.8% and 10.8% of accidents happen near traffic signals and crossings, respectively. The severity of accidents is lower, on average, when they occur near both traffic signals and crosswalks than when they occur near only one of these two features. Accidents near crossings and stations have among the lowest mean severity and distance affected of all those recorded.

One troubling possibility is that the low severity and distance affected might reflect the involvement of pedestrians and bicyclists in a plurality of accidents near these infrastructure features; pedestrians and bicyclists are easily removed from the road after accidents, and so delays may be relatively

small. This interpretation is perhaps challenged, but also perhaps supported by the fact that 43% of accidents happening near public transit stations also happen near crossings. On the one hand, this points to pedestrian involvement being a significant part of accidents near transit stations. In fact, the mean severity of accidents near stations but not near crossings is higher than accidents near both, which may suggest that pedestrian involvement lowers severity, when in fact pedestrian involvement is actually likely to increase risk of injury and death (to the pedestrian, of course). On the other hand, accidents near crossings and not stations are likewise more severe than those near both, though not as severe as accidents near stations and not crossings; this goes against intuition, which would suggest that severity would be more similar between accidents happening near crosswalks or not near crosswalks in the vicinity of stations than between accidents happening near crosswalks around stations and not around stations, since accidents near stations would have much in common, and accidents happening near crosswalks either near stations or far from stations would have less in common. As we discuss in our ethics section, the possibility of a bias against the welfare of pedestrians calls into question the appropriateness of the characterization of severity espoused in this dataset.

5. ETHICAL IMPLICATIONS AND CONCLUSIONS

We must consider how others might act on our conclusions, and how our assumptions and our limitations of our data might affect it. The measure of accident severity used in the dataset accounts only for the delay to traffic—a fact not conveyed by the data itself or by either of the papers supporting the dataset, but simply by the column description provided on kaggle—and breaks it up into 4 categories rather than providing more specific numbers. The only other measure we have is the length of road impacted by an accident. For this reason, our model is very likely to undervalue the severity of accidents that have a minimal impact on traffic. For instance, severe or even fatal accidents involving pedestrians, bicyclists, or motorcyclists are unlikely to leave wreckage that might seriously impede traffic and might be considered minor by our model. On the other hand, if a semi-truck were to overturn and spill its contents onto the road, even without serious injury, traffic would be severely impeded, and the accident would be considered severe. That traffic-flow should be our primary measure of severity is troubling. Though impeded traffic can cause serious problems, especially for emergency vehicles, it must not be considered more troubling than actual injury or death. Simplicity is its failing. A measure of severity that is easily quantified and compared may fail to capture important information. Of course, more comprehensive metrics that would include counts of injuries and fatalities, medical costs, and long-term impacts on the lives of those involved may both increase the complexity of our severity ratings more than is reasonable and risk violating the privacy of those involved. Regardless, a

clear disclosure of the precise metrics used to measure severity is a reasonable minimum expectation, as it is necessary if we are properly to account for the biases introduced by the metrics used. The use of a term like “severity” to describe only a single, relatively minor component of the overall severity of an accident is unacceptable. Such naming conventions are likely to lead to inappropriate assumptions and bad conclusions at best.

If we had more time, we might have parsed the street-name associated with each accident to determine whether it is a highway, a country road, a city street, or some other kind of road. With this, we could compare these different varieties of road and determine how they relate to crash severity. Additionally, we might have taken random samples of the data such that we had equal numbers of crashes near to each of the infrastructure features, so that we could more fairly compare the importance of each feature in predicting either of our measures of crash severity.

REFERENCES

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